

SmartReport User Manual

1 Introduction

Introduction

SmartReport™ is a comprehensive nuclear cardiology reporting program developed by Dr. Larry Klein of POSMedIC LLC and Syntermed. Syntermed, Inc. serves as the distributor of the *SmartReport*™ application. *SmartReport*™ can be used as either a stand alone application or as a totally integrated package with the Emory Cardiac Toolbox (ECTb™) application.

The *SmartReport*™ program provides a simple method for the physician to quickly and efficiently generate a nuclear cardiology report of the patient's myocardial perfusion and function. Options are available to generate a normal or abnormal report either with a few clicks or by using the comprehensive tools provided in *SmartReport*™. The phrases and structure in SmartReport were compiled based on the ASNC Imaging Guidelines for Nuclear Cardiology Procedures - Standardized reporting of radionuclide myocardial perfusion and function (<http://www.asnc.org/imageuploads/ImagingGuidelinesReportingJuly2009.pdf>). User interaction involves selecting and entering comments and phrases from drop down boxes and entering text in edit boxes located within the various *SmartReport*™ menu screens. This information is ultimately assimilated to generate the final patient's report. The final report offers flexibility to be modified to an output specified by the user. *SmartReport*™ executes on Mac OS, and Windows XP/7/8, system platforms.

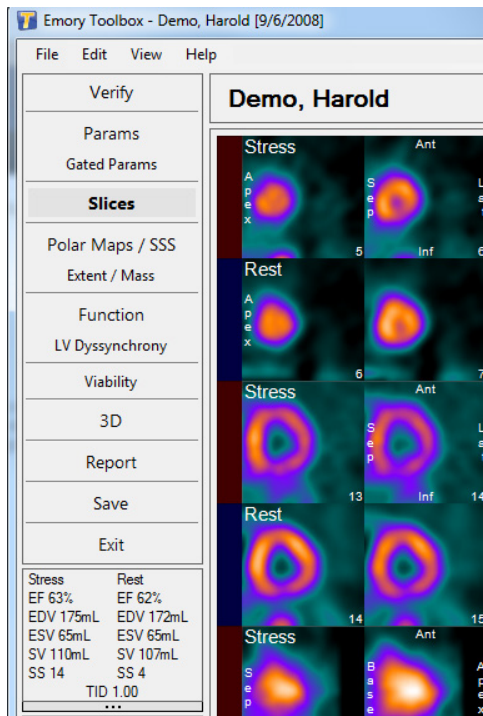
ECTb™ Integration - The *SmartReport*™ program can be obtained as a fully integrated application within the ECTb™ nuclear cardiology analysis program. The ECTb™ program provides a method to automatically interpret the SPECT or PET nuclear cardiology study using an image support system (IDS) developed by Dr. Ernest Garcia at Emory University. Syntermed's Image Decision Support (IDS) program is a plugin for the 2nd generation application architecture that works directly with Emory Toolbox 4.0. IDS extracts regional quantitative parameters of myocardial perfusion, reversibility, viability and function during the processing of the SPECT and PET studies within the Emory Toolbox 4.0. These regional parameters are converted to probability of abnormality (certainty factors) and submitted to a decision support algorithm that is updated with the latest scientific/clinical knowledge to reach an impression of the patient's heart status. The IDS expert system is comprised of the knowledge base, the inference engine and the justification engine. The knowledge base consists of heuristic rules derived from domain experts. The conclusions reached by the IDS and justifications for each conclusion are used to automatically generate a structured report (SmartReport) for the diagnostician to review, learn from the justifications, and either modify and/or approve. This configuration also provides the capability of incorporating images and data generated by the ECTb™ application directly into the *SmartReport*™ report.

SmartReport™ provides a cloud based, Client/Server application. This means that separate *SmartReport*™ clients can be located at the front desk, EKG room, and imaging room where data related to patient demographics, EKG, and imaging protocol can be entered by appropriate medical personnel eliminating the need for entry of this information by the physician, thereby reducing the time the physician needs to complete the report.

1.1 Launch SmartReport

Launching SmartReport - Integrated with Emory Toolbox

To launch SmartReport from the Emory Toolbox you will need to click on the Report button which is shown on the navigation bar located on the left side of the display. The quantitative information from the study will then be sent to the IDS automatic interpretation system and these results will then be sent to SmartReport prior to launching the SmartReport application.

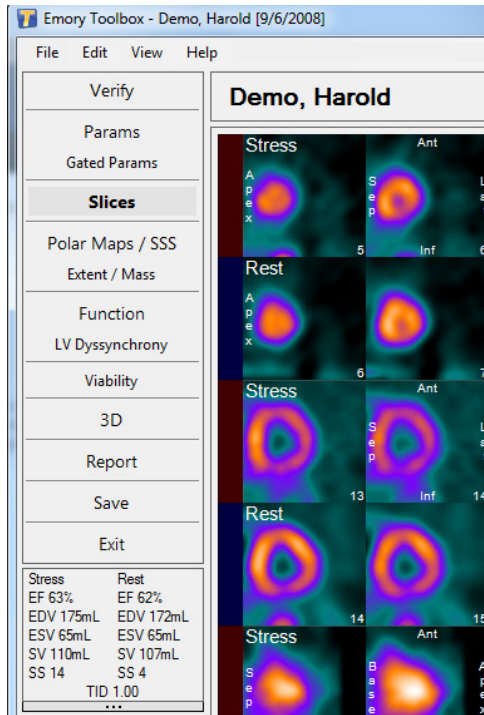


1.2 Setting Options

Setting Options

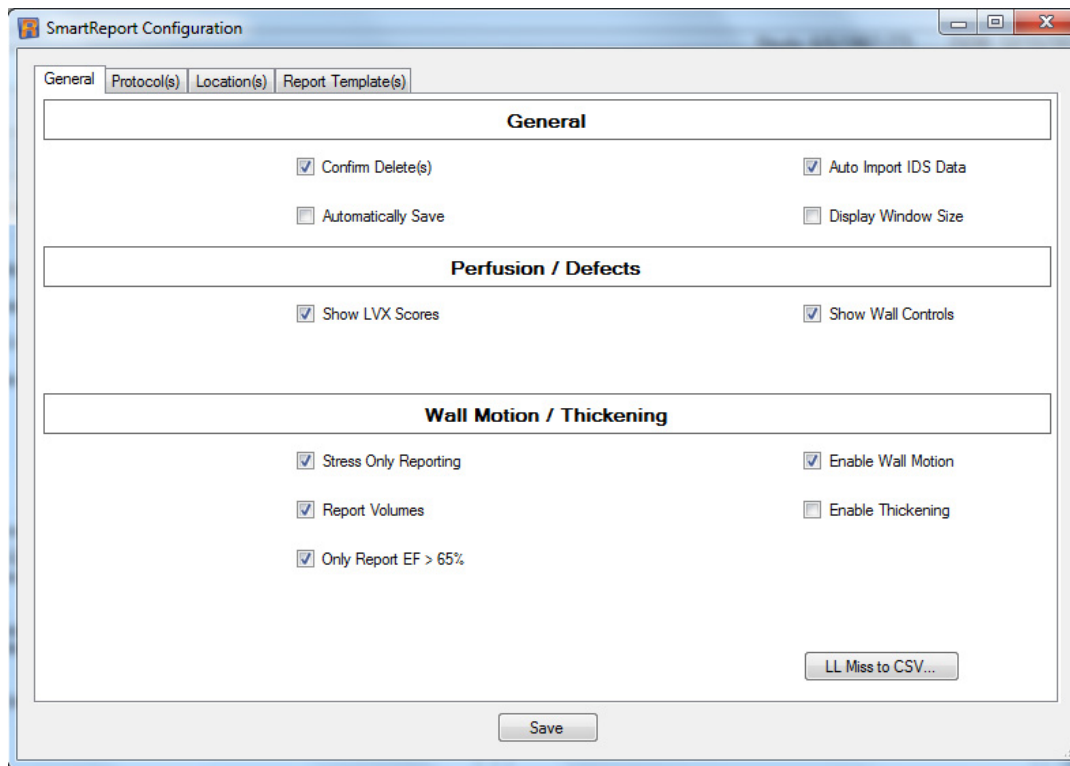
SmartReport has been developed to adhere to the reporting standards established by the American Society of Nuclear Cardiology (ASNC). However, SmartReport does provide the ability to customize any of the drop down lists, report templates, or phrases within the application. Since the program is based on the ASNC standards we recommend that new phrases be added and the existing phrases left as they are currently listed. Changes can be entered for 6 categories: General, Personnel, Protocol(s), Locations(s), Report Template(s), and Lookup(s).

The changes to SmartReport are accomplished by going to the Edit/Options screen which is shown below.



1.2.1 General

General



General

- Confirm Delete - if checked this option will present a dialog window when a study is going to be deleted.
- Auto Import IDS Data - if checked the study interpretation generated by IDS in the Emory Toolbox will be imported into the perfusion window in a natural language format.
- Display Window Size - if checked the existing window size will be displayed.
- Automatically Save - save the report automatically if checked.

Perfusion / Defects

- Show LVX Scores - if checked the automatic scores derived from IDS will be displayed on the polar maps.
- Show Wall Controls - if checked the bars signifying the 7 walls of the heart around the polar map will be displayed and are active. This allows the user to set all segments pertaining to a wall as abnormal with a single click.

Wall Motion / Thickening

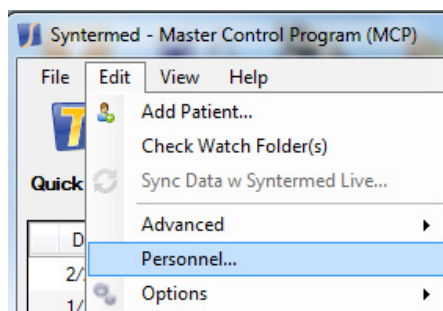
- Stress Only Reporting - if checked only the Stress study will be enabled on the screen for reporting.
- Report Volumes - if checked the volumes generated by the Emory Toolbox will be displayed.
- Only Report EF > 65% - if checked only the EFs > 65 % will be listed on the report.
- Enable Wall Motion - if checked it will enable the wall motion polar maps so that the scores can be entered manually.
- Enable Thickening - if checked it will enable the scores to be changed manually.

LL Miss to CSV... -

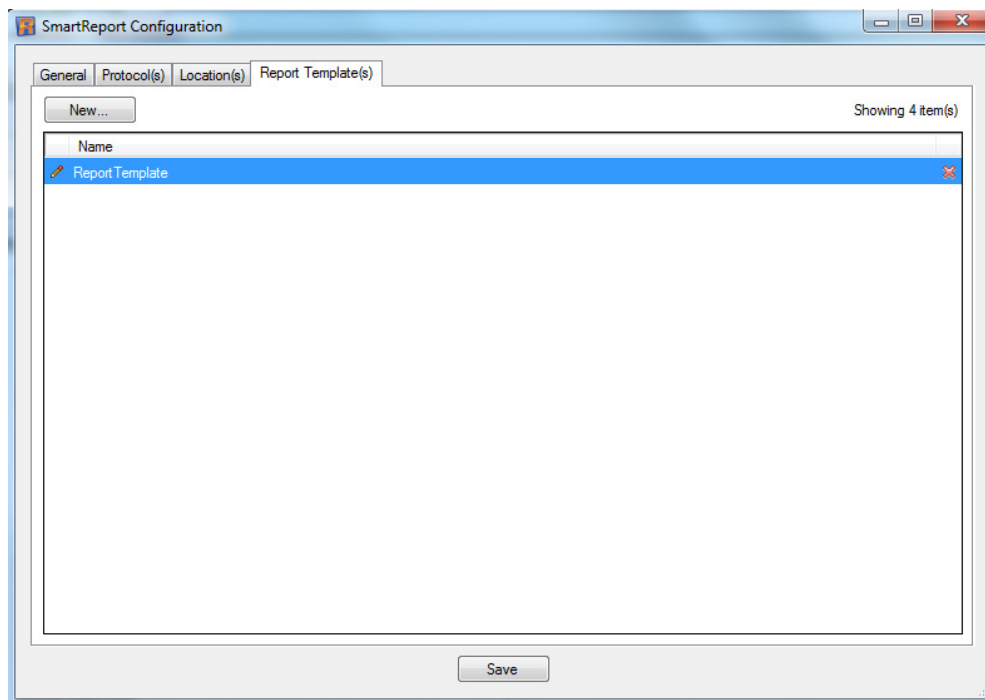
1.2.2 Personnel

Personnel

The user can manage personnel that will be associated with the patients studies at the institution by navigating to the Personnel tab in the Options dialog which can be accessed from the MCP application as shown below:



The main personnel screen listing all of the personnel is shown below.



If you want to edit an existing entry click on the pencil symbol. If you want to add new personnel then click on the New button and the following screen will be displayed.

The new personnel information should be added on this screen. If you want to use a digital signature then click within the Signature box and you will be prompted to locate the digitized signature file on your system. Once all of the information for the person is entered click the Save button.

Roles

In the dialog above, the personnel entry should be added to all applicable roles that the personnel will hold at the institution.

The list of Roles can be modified by changing the items listed in the "NuclearPerfusionPersonnelRoles" Lookup.

- The entries with "Enum Name" set to the following cannot be removed, but their "Display Name" can be changed:
 - PrimaryAttending
 - StressAttending

When the following tags are used in the report template, the information will be rendered as follows in the report.

[-Personnel-]

- Entries will be placed on lines based on category divided into the following groups:
 - If the "Enum" value of a Role contains "Attending"
 - If the "Enum" value of a Role contains "Referring",
 - All other entries.
- Entries will be grouped together by the "Display Name" of the Role.
- If a line does not have any valid entries, it will be left blank.
- The Referrings line will always be last.

[-ReviewerName-]

- This will display the name of the Personnel selected in the PrimaryAttending Role. If none has been selected, it will be left blank.

[-StressSupervisorName-]

- This will display the name of the Personnel selected in the StressAttending Role. If none has been selected, it will be left blank.

1.2.3 Protocol(s)

Protocol

The protocol option screen allows the user to either edit existing protocols or to add new ones for both Nuclear and Stress protocols. All of the Nuclear and Stress protocols currently existing in the SmartReport database are listed on this screen.

The order in which the protocols show up in their respective lists in SmartReport can be modified using the up/down arrows for each list. **Please remember to hit the Save button before leaving this dialog.**

The SmartReport Configuration dialog box has tabs for General, Protocol(s), Location(s), and Report Template(s). The Protocol(s) tab is active, showing two lists: Nuclear Protocol(s) and Stress Protocol(s). Each list has a 'New...' button and a 'Reset' button. The Nuclear list contains 18 protocols, each with a pencil icon for editing. The Stress list contains 8 protocols, also with pencil icons. Arrows between the lists allow moving protocols from one to the other. A 'Save' button is at the bottom.

Nuclear Protocol(s)	
Name	
One-Day Rest/Stress Tc-99M Sestamibi	✖
Two-Day Rest/Stress Tc-99M Sestamibi	✖
One-Day Rest/Stress Tc-99M Sestamibi (SPECT / CT)	✖
Two-Day Rest/Stress Tc-99M Sestamibi (SPECT / CT)	✖
Rest Tc-99M Sestamibi	✖
Stress Tc-99M Sestamibi	✖
Rest/Stress Rb	✖
Rest/Stress NH3	✖
Rest Rb	✖
Rest Rb / FDG	✖
Rest / Stress Rb / FDG	✖
One-Day Rest/Stress Tc-99M Tetrofosmin	✖
One-Day Stress/Rest Tc-99M Tetrofosmin	✖
Two-Day Rest/Stress Tc-99M Tetrofosmin	✖
One-Day Rest/Stress Tc-99M Tetrofosmin (SPECT / CT)	✖

Stress Protocol(s)	
Name	
Bruce	✖
Dobutamine	✖
Regadenoson	✖
Adenosine	✖
Bruce (converted to Dobutamine)	✖
Bruce (converted to Regadenoson)	✖
Dipyridamole	✖
Rest Only	✖

Nuclear Protocol(s)

If you wish to create a new protocol then click on the "New..." button. To edit a nuclear protocol, click on the pencil icon. Fill out the available fields as described below and hit the "Save" button.

The Add/Edit Nuclear Protocol dialog box contains the following fields: Display Name (text box), Enum Name (text box), Modality (dropdown menu set to SPECT), Acquisition(s) (table with columns Acquisition and Radiopharmaceutical), Type (dropdown menu set to Rest), Radiopharmaceutical (dropdown menu set to Tc-99m Sestamibi), AC Type (dropdown menu set to None), Multiple Day(s) (checkbox), Phrase (text area), Visible (checkbox checked), Save button, and Cancel button. A 'Reset Phrase Text' link is also present.

Acquisition	Radiopharmaceutical

Display Name

This is the exact text that will be displayed in the list of Protocols in SmartReport.

Enum Name

This is an internal use, automated, required field that is a specially formatted representation of the protocol name. This value cannot contain spaces or special characters (/ , | , & , * , etc...) and cannot start with a number. This value should only be used once per Site.

Modality

The default modalities are shown above, but can be modified by editing the "Modalities" lookup via the Lookups tab. You may need to save and reopen the SmartReport options dialog for the changes to take effect.

Acquisition Type(s)

The default acquisition types are shown above, but can be modified by editing the "AcquisitionTypes" lookup via the Lookups tab. You may need to save and reopen the SmartReport options dialog for the changes to take effect. More than one can be selected.

Radiopharmaceutical

This drop down allows the user to select the radiopharmaceutical to be used in the acquisition.

AC Type

Allows the user to select which type of attenuation correction was used: Transmission or computed tomography (CT)

Multiple Day(s)

This field is not supported at this time.

Phrase

This is the phrase that will be filled in based on the values selected in SmartReport related to this protocol. For simplicity, please copy/paste the phrase from a similar, pre-existing protocol and modify it as necessary.

[-TagName-]

All items in this format will be replaced with values from the SmartReport interface. The appropriate values for each acquisition should be configured by replacing the keyword (in red) like in the example tags listed below:

[-NuclearProtocolDose**Stress**-]

[-NuclearProtocolDose**Rest**-]

[-NuclearProtocolDose**FDG**-]

Other available tags are as follows:

[-StressProtocolSummary-]

[!-----INSTRUCTIONS-----!]

All items in this format should be modified at this time with the appropriate text.

All other text will be displayed as is in the report.

Visible

The protocol will be hidden from list in SmartReport when unchecked.

Stress Protocol(s)

If you wish to create a new protocol then click on the "New..." button. To edit a nuclear protocol, click on the pencil icon. Fill out the available fields as described below and hit the "Save" button.

The screenshot shows a Windows-style dialog box titled "Add/Edit Stress Protocol". It contains several input fields and checkboxes. At the top are "Display Name" and "Enum Name" text boxes. Below them is an "Exercise" checkbox and an "Exercise Type" dropdown menu. Under "Exercise Type" is a "Pharmacological" checkbox, which when checked, reveals a list of pharmacological agents: Dobutamine, Regadenoson, Dipyridamole, Atropine, Adenosine, and Aminophylline, each with its own checkbox. Below this is a large text area for "Phrase". At the bottom left is a "Visible" checkbox, and at the bottom right are "Save" and "Cancel" buttons.

Display Name

This is the exact text that will be displayed in the list of Protocols in SmartReport.

Enum Name

This is an internal use, automated, required field that is a specially formatted representation of the protocol name. This value cannot contain spaces or special characters (/ , | , & , * , etc...) and cannot start with a number. This value should only be used once per Site.

Exercise

Check this box if the protocol has an exercise component.

Exercise Type

Select the exercise type from the drop down list.

Pharmacological

Check this box if the protocol has an pharmacological component.

Pharmacological Agent(s)

The default pharmacological agents are shown above, but can be modified by editing the "PharmacologicalStressAgents" lookup via the Lookups tab. You may need to save and reopen the SmartReport options dialog for the changes to take effect. More than one can be selected.

Phrase

This is the phrase that will be filled in based on the values selected in SmartReport related to this protocol. For simplicity, please copy/paste the phrase from a similar, pre-existing protocol and modify it as necessary.

[-TagName-]

All items in this format will be replaced with values from the SmartReport interface. The appropriate values for each acquisition should be configured by replacing the keyword (in red) like in the example tags listed below:

[-PharmaStressDose**Dobutamine**-]
[-PharmaStressDose**Atropine**-]

Other available tags are as follows:

[-ExerciseDuration-]
[-StressProtocol-]
[-ReasonForExerciseTermination-]
[-ReasonExerciseCouldNotBeAttempted-]
[-ReasonForStressProtocolConversion-]

[!-----INSTRUCTIONS-----!]

All items in this format should be modified at this time with the appropriate text.

All other text will be displayed as is in the report.

Visible

The protocol will be hidden from list in SmartReport when unchecked.

Please remember to hit the Save button before leaving this dialog.

1.2.4 Location(s)

Location

The location option allows the user to set the name of the institution, address, phone number, etc.. The Location screen is shown below.

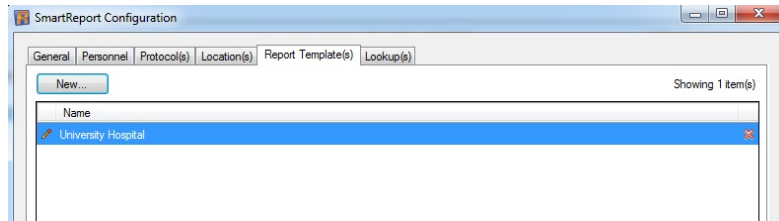
The screenshot displays the 'SmartReport Configuration' application window. The 'Location(s)' tab is selected, showing a table with columns for Name, Room, and Camera. A single entry, 'University Hospital', is listed. An 'Add/Edit Location' dialog box is open in the foreground, containing the following fields: Name (text box), Address (text box), Phone (text box), Room (text box), Camera (text box), Description (text box), and Report Template (dropdown menu). A 'Save' button is at the bottom of the dialog, and a 'clear' link is next to the Report Template dropdown. The background application window also has a 'Save' button at the bottom.

All of the current locations will be listed on the screen and they can be edited by clicking on the pencil symbol. If you wish to enter a new location then click on the New button and the dialog shown above will be shown. Enter the information for your institution on this screen and then click on the Save button.

1.2.5 Report Templates

Report Template(s)

The report template can be generated with this option and this screen is shown below.



The existing templates will be shown and can be edited by clicking on the pencil symbol. If you are creating a new template then click on the New button and the following screen will be displayed.

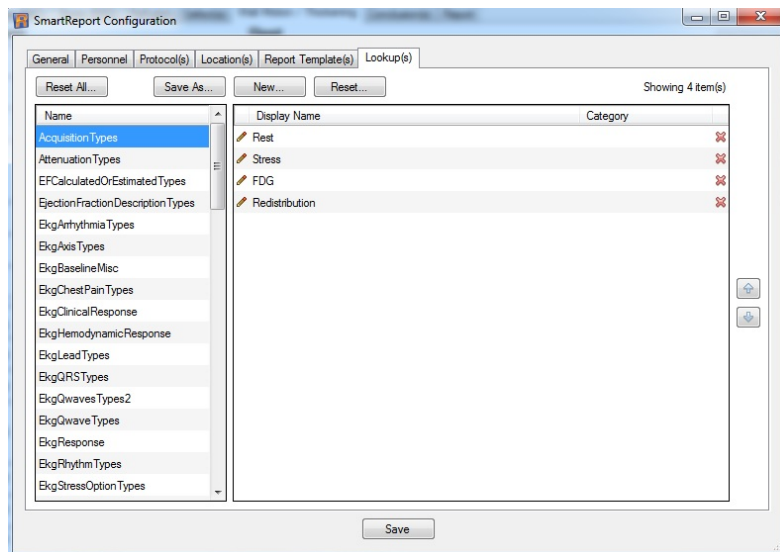
The image shows the 'Add/Edit Template' window. It has two main sections. The top section, 'Template Info', contains fields for 'Name', 'Default Protocol', 'Default Stress Protocol', 'Default Location', and 'Header Logo' (with a 'clear' link). A 'Save' button is at the bottom right of this section. To the right of these fields is a table for selecting roles and their names. The bottom section is a rich text editor with a toolbar (Font, Bold, Italic, Underline, etc.) and a text area containing placeholder text for a report template. The text includes location details, patient information, exam details, clinical findings, and protocol information, all enclosed in brackets to denote variables. A 'Default Text' button is at the bottom left, and checkboxes for 'Header/Footer' and 'Legend' are at the bottom right.

The new template can be generated from this screen. The name of the template is added first and then various defaults for protocols, location, hospital header, and physicians can be added. Any text within [] means that it is a variable and cannot be changed. The rest of the text can be changed and moved around to customize the template to the preference of the user. Once you are finished with the template click the Save button and it will now be available within SmartReport.

1.2.6 Lookup(s)

Lookup(s)

The Lookup option allows the user to customize any of the lists contained within SmartReport. The Lookup screen is shown below.

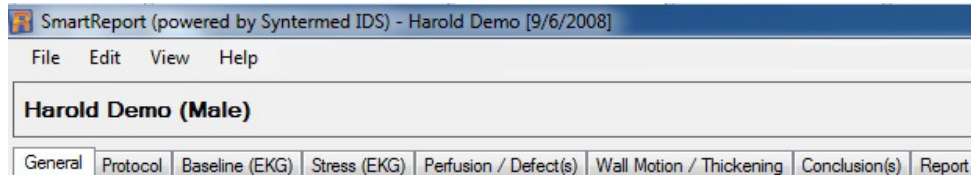


The list of all of the Lookup tables are shown on the left column and the items associated with that Lookup table is on the right hand side. You can edit any of the existing items by clicking on the pencil symbol and if you want to add a new item click on the New button. Once the items are entered or changed click on the Save button and the new lookup table will be entered into the SmartReport database.

2 Report Screens

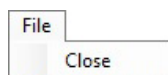
Report Screens

There are 8 screens used to complete a patients nuclear cardiology report which are described below. The screens provide a combination of lists and drop down boxes containing phrases that are selected to generate the patients report. The phrases and structure in SmartReport were compiled based on the ASNC Imaging Guidelines for Nuclear Cardiology Procedures - Standardized reporting of radionuclide myocardial perfusion and function (<http://www.asnc.org/imageuploads/ImagingGuidelinesReportingJuly2009.pdf>). The user is allowed to customize the reporting system using edit, add, or delete options for any of the phrases, lists, or statements in SmartReport and to customize the reporting templates. The process for customization of SmartReport is covered in section 1.2 Setting Options. The SmartReport screen is shown below.

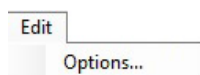


There are 4 options on the top bar: File, Edit, View, and Help

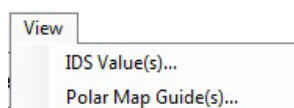
File - This option currently provides an option to Close the application.



Edit - This option allows for selecting Options for setting the preferences the user wants to use for SmartReport. See section 1.2 Setting options for a discussion of how these values are set.



View - This option allows for displaying the values used for the IDS automatic interpretation algorithm and for displaying the Polar Map Guide.



IDS Value(s) - this screen is shown below.

IDS Data Review

Property Name	Value
LvSize	Mildly Dilated
LvPerfusionSummary	Abnormal
LvPerfusionSummaryClassification	Ischemia
CoronaryDistribution	LADAndLCxAndRCA
LvFunctionSummary	Abnormal
GlobalLvFunction	Normal
ScanSignificance	HighRisk

Defect(s)

Defect(s)	Property Name	Value
1	Size	Moderate
2	Severity	Mild
3	Type	Mixed
	PerfusionDefect Summary	Ischemia
	PerfusionDefect SummaryArtifa...	
	RegionalFunctionalSummary	InconsistentWithPerfusion
	CoronaryDistribution	LAD
	ExtraCardiacActivity	None

Segment Name	ScoringCode
BasalAnterior	F
BasalAnteroseptal	F
MidAnterior	NC
ApicalAnterior	M

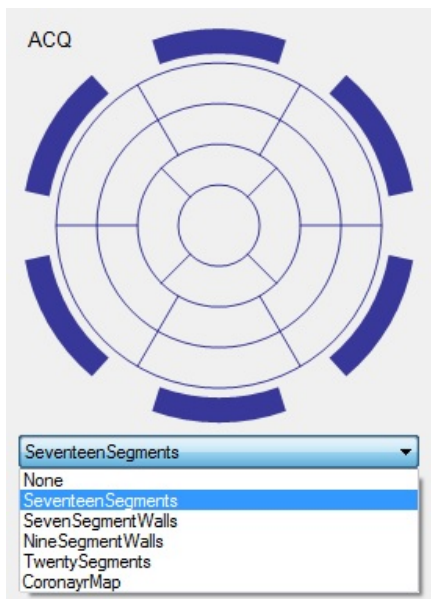
Score(s)

Study Type: Perfusion

Score Type: Rest

Segment Name	Value
BasalAnterior	3
BasalAnteroseptal	3
BasalInferoseptal	3
BasalInferior	3
BasalInferolateral	2
BasalAnterolateral	1
MidAnterior	0
MidAnteroseptal	0
MidInferoseptal	0
MidInferior	2
MidInferolateral	1
MidAnterolateral	0
ApicalAnterior	1
ApicalSeptal	0
ApicalInferior	2
ApicalLateral	0
Apical	0

Polar Map Guide - this screen is shown below and represents the polar map diagram with a drop down list which demonstrate the various configurations for the polar map.



Help - this option is used to display the User Manual.

The 8 screens that are used to create the nuclear report are described below.

1. General
 - The General screen is used to enter patient demographics, indications, risk factors, medications, past cardiac procedures, physicians, height, weight, etc.. The patient demographic information is imported from the Emory Toolbox and the rest of the page can be filled out by the secretary or technologist prior to imaging the patient.
2. Protocol
 - The Protocol screen contains information related to the imaging and stress protocol used, doses injected, level of exercise, hemodynamics, and technical quality of the study.
3. Baseline (EKG)
 - This screen allows the user to enter the results of the patients baseline EKG.
4. Stress (EKG)
 - This screen allows the user to enter the results of the patients stress EKG
5. Perfusion / Defect(s)
 - The results of the patients perfusion defect is entered on this screen. If SmartReport is integrated with the Emory Toolbox then the automatic interpretation generated by IDS is imported and populated on this page in SmartReport.
6. Wall Motion / Thickening
 - The Wall Motion / Thickening page allows the user to enter the wall motion interpretation for the 17 walls of the heart. The thickening information is automatically populated from the IDS output if SmartReport is integrated with the Emory Toolbox.
7. Conclusion(s)

- Conclusion statements related to what the user wants to enter into the report for the patients is completed on this screen.
8. Report
- The final report is displayed containing all of the values entered in the previous 7 screens.

2.1 General

General Screen

There are 4 areas which need to be completed on this screen: Height and Weight, Personnel information, Indications, Risk Factors, and Past Cardiac Procedures (if any). The patient demographics (name, number, dob, study date, etc.) have previously been entered from the MCP program. The areas which require user interaction are discussed below.

Height and Weight - The patients height in inches and weight in pounds should be entered into the appropriate edit boxes. The height and weight are then converted to centimeters and kilograms automatically. The body mass index (BMI) and body surface area (BSA) is then automatically calculated from the height and weight information.

Personnel Information - The names of the physicians involved in the study can be entered here from drop down lists. If the physician is not in the database then they can be added by clicking on the + symbol. When this is clicked the personnel search screen will be displayed and is shown below.

First choose the desired Role for the new Personnel to hold in the report, then choose the appropriate entry. The user has the ability to add a Personnel entry at this time via the "New..." button if necessary.

Note: There can only be one Personnel entry per report for PrimaryAttending and StressAttending.

Note: If no StressAttending is selected, but the report template uses related tags, the selected entry for PrimaryAttending will be displayed automatically in its place.

Indications - The Indications box presents a list of indications which can be selected by clicking on the indication which will then show up in the Indications box at the bottom of the screen. Everything listed in the boxes at the bottom indicate information that is going to be entered into the patients report. If you would like to enter different indications then this can be done by going to Edit/Options/Lookups and then select Indications from the list in the left column. When this is done the screen below will be shown. You can then delete or add new indications to the list.

SmartReport Configuration

General | Personnel | Protocol(s) | Location(s) | Report Template(s) | Lookup(s)

Reset All... Save As... New... Reset...

Showing 25 item(s)

Name	Display Name	Category
EkgSTSegmentTypes	Chest Pain	
ExerciseTerminationReasons	Chest Discomfort	
ExtraCardiacActivityTypes	Abnormal EKG	
GeneralConclusions	Dyspnea	
GlobalLvFunctionSummaryTypes	S/P CABG	
GlobalWallMotionTypes	S/P MI	
Indications	S/P PCI	
LungUptakeTypes	Poor Functional Status	
Lv17Segments	End Stage Renal Disease	
LvCoronaries	Elevated Cardiac Biomarkers	
LvDefectCauseAndOrTypes	Pre-Op Evaluation	
LvDefectCauseTypes	Pre-Op Evaluation With Poor Functional Status	
LvDefectProbabilityOfCauseTypes	Pre-Op Evaluation For Renal Transplant	
LvDefectProbabilityOfCauseTypes2	S/P Heart Transplant	
LvDefectScoringCodes	Chronic Renal Failure On Dialysis	
LvDefectSeverities	Pre-Op Evaluation Liver Transplant	
LvDefectSizes	Heart Failure	

Save

Risk Factors - The Risk Factors box presents a list of risk factors which can be selected by clicking on the risk factor which will then show up in the Risk Factors box at the bottom of the screen. Everything listed in the boxes at the bottom indicate information that is going to be entered into the patients report. If you would like to enter different risk factors then this can be done by going to Edit/Options/Lookups and then select Risk Factors from the list in the left column. When this is done a screen similar to the Indications screen shown above will be displayed. You can then delete or add new risk factors to the list.

Past Cardiac Procedures/Other Conditions/Medications - If the patients had any previous cardiac procedures, other conditions or medications that you would like listed in the report then these items can be selected from the related screens. If you would like to enter different information for any of these areas then this can be done by going to Edit/Options/Lookups and then select Medical History Procedures, Other Conditions, or Patient Medications from the list in the left column. When this is done a screen similar to the Indications screen shown above will be displayed. You can then delete or add new entries to the list to the list.

2.2 Protocol

Protocol

The protocol screen is used to enter the nuclear and stress protocol used for the acquisition and stress of the patient. The protocol screen is shown below.

SmartReport (powered by Syntermmed IDS) - Harold Demo [9/6/2008]

File Edit View Help

Harold Demo (Male) Study Date: 9/6/2008 DOB: 10/17/1956 (57) ID: N/A

General | Protocol | Baseline (EKG) | Stress (EKG) | Perfusion / Defect(s) | Wall Motion / Thickening | Conclusion(s) | Report

Nuclear Protocol

- One-Day Rest/Stress Tc-99m Sestamibi
- One-Day Rest/Stress Tc-99m Tetrofosmin
- Rest/Stress Rb
- Rest Tl-201/Stress Tc-99m Sestamibi
- Rest Tl-201/Stress Tc-99m Tetrofosmin
- Two-Day Rest/Stress Tc-99m Sestamibi
- Two-Day Rest/Stress Tc-99m Tetrofosmin
- Rest Rb / FDG
- Rest / Stress Rb / FDG
- Rest Tc-99m Tetrofosmin
- Stress Tc-99m Tetrofosmin
- One-Day Stress/Rest Tc-99m Tetrofosmin
- Rest Tc-99m Sestamibi
- Rest Tl-201
- Two Day Rest Tl-201/Stress - Tc-99m Sestamibi
- Stress/Redistribution Tl-201
- Rest Redistribution Reinjection Thallium
- Stress/Redistribution Tl-201 With Reinjection
- Rest Rb
- One-Day Rest/Stress Tc-99m Tetrofosmin(SPECT / CT)
- Two-Day Rest/Stress Tc-99m Tetrofosmin(SPECT / CT)
- Stress Tc-99m Sestamibi

Stress Protocol

- Bruce
- Dobutamine
- Regadenoson
- Adenosine
- Bruce (converted to Dobutamine)
- Bruce (converted to Regadenoson)

☐ Normal Study *

Nuclear Doses

Rest: 10
Stress: 30

Exercise

Mins: 9 Secs: 22 Stage: 3 METS: 10

Reason(s) for Exercise Termination: Target Heart Rate Achieved

Hemodynamics

	BP	HR
Baseline	117 / 76	65
Peak	125 / 85	113
THR	143	PMHR 169 % Pred 67%

Technical Quality

Overall Quality: Good

Motion: ☐ Motion Corrected

Adjacent Gut Activity: ☐ Attenuation Corrected

Lung Uptake: ☐ Attenuation Corrected

Attenuation Type(s): Mild Breast, Moderate Breast, Marked Breast, Mild Diaphragmatic, Moderate Diaphragmatic, Marked Diaphragmatic, Arm Imaged at Patient's Side, Pacemaker

Pharmacological Stress

Dose(s): N/A

Comments

Protocol Phrase Preview

The patient was brought to the nuclear laboratory and given 10.0 mCi of Tc-99m sestamibi intravenously. Approximately 15 minutes later, the patient had the resting images performed. The patient exercised under the supervision of a physician for a total of 9 min(s), 22 sec(s) achieving stage 3 (10.0 METS) of the Bruce protocol. Exercise was terminated because the target heart rate was achieved. After the IV administration of 30.0 mCi of Tc-99m sestamibi at peak exercise, stress tomographic images of the heart were obtained.

Location / Facility: Save Save and Close Report Template

Nuclear Protocol - Most of the nuclear cardiology protocols used in nuclear labs for imaging patients are listed in the Nuclear Protocol box. Click on the protocol which you are using for the patient and the text that is used for protocol will be listed in the Protocol Phrase Preview box. Any of these

protocol names or text used can be edited and this is accomplished by going to Edit/Options/Protocol(s). Instructions for editing the protocols is explained in section 1.2 Setting Options.

Nuclear Doses - the technologist needs to enter the rest and stress doses in the nuclear doses box. If the doses are not entered then when the final report is generated it will notify the physician that the doses were not entered since this is a Required variable.

Normal - the normal check box is used if the study is normal. This will automatically put in preset phrases for nuclear perfusion, wall motion, and conclusion. These phrases can be edited by going to Edit/Options/Lookups and selecting NormalPhrases and Normal/AbnormalConclusions from the left hand list and then selecting the item that will be edited located in the right hand list. Instructions for editing the protocols is explained in section 1.2 Setting Options.

Stress Protocol - Most of the stress protocols used in nuclear labs for stressing patients are listed in the Stress Protocol box. Click on the protocol which you are using for the patient and the text that is used for protocol will be listed in the Protocol Phrase Preview box. If it is a Bruce protocol then the Exercise boxes will be enabled and if it is pharmacological stress then the Pharmacological Stress box will be enabled. The Hemodynamics are entered for both methods of stress in the Hemodynamics boxes. Any of these protocol names or text used can be edited and this is accomplished by going to Edit/Options/Protocol(s). Instructions for editing the protocols is explained in section 1.2 Setting Options.

Technical Quality - Entries related to the technical quality of the study are entered in this section. These have to do with TID, overall quality, motion, adjacent gut activity, lung uptake and attenuation.

Comments - the user can enter and comments into this box and it will be displayed in the final report in the Protocol section of the report.

Location/Template Bar

The bottom of each screen contains a method that allows the user to select the location and template that are to be used for the study. The bar is shown below.



The screenshot shows a horizontal bar with two dropdown menus and two buttons. The first dropdown menu is labeled 'Location / Facility' and has 'University Hospital' selected. The second dropdown menu is labeled 'Report Template' and also has 'University Hospital' selected. Between the two dropdown menus are two buttons: 'Save' and 'Save and Close'.

Location / Facility - is used to list the name, address, phone, etc. for the facility. The method for generation of the Location and Facility information is described in section 1.2 Setting Options.

Save - is used to save the information entered in the particular screen. Note: this is done automatically once you leave a screen.

Save and Close - this is used if you want to close out of SmartReport before the report is completed.

Report Template - these are the report templates set up by the site which is accomplished by going to Edit/Options/Report Template(s). The method for generation of the report templates is described in section 1.2 Setting Options.

2.3 Baseline (EKG)

Baseline (EKG)

This screen allows the user to enter information related to the results of the patients baseline EKG. This screen is shown below. Information can be entered related to Rhythm, Axis, QRS, Q Waves, ST Segment(s), Misc, ST Segment Changes and Arrhythmia(s). The information selected from these sections will be displayed in the left and right hand boxes at the bottom of the screen and this is the information that will be entered into the final report. Any of the phrases listed on this page can be edited or new phrases created by going to Edit/Options.Lookup(s) and go to the particular item listed on the left hand column, then edit the particular entry listed on the right hand column. This editing process is described in section 1.2 Setting Options.

SmartReport (powered by Syntermmed IDS) - Harold Demo [9/6/2008]

File Edit View Help

Harold Demo (Male) Study Date: 9/6/2008 DOB: 10/17/1956 (57) ID: N/A

General Protocol Baseline (EKG) Stress (EKG) Perfusion / Defect(s) Wall Motion / Thickening Conclusion(s) Report

Baseline

Rhythm (choose one)

Normal Sinus Rhythm
Sinus Bradycardia
Sinus Tachycardia
1st Degree AV Block
Atrial Fibrillation
Atrial Flutter
Ectopic Atrial
NSR W/ Sinus Arrhythmia
NSR W/ Wenckebach
NSR W/ 2nd Degree AV Block
Junctional Rhythm

Axis (choose one)

Normal Axis
Leftward Axis
Left Axis Deviation
LAFB
Rightward Axis
Right Axis Deviation
LFTB
Indeterminate Axis

QRS (choose one)

LB88
RB88
Incomplete LB88
Incomplete RB88
VCD
LVH With Repolarization

Q Wave(s)

Anterior
Septal
Lateral
Posterior

ST Segment(s)

Non-Specific ST T Changes
Non-Specific T Changes
Non-Specific T Changes
Early Repolarization
Diffuse ST Elevation
NSSTT Consider Inferior Ischemia
NSSTT Consider Anterior Ischemia
NSSTT Consider Anterior Ischemia
Anterior ST Elevation Possibly Acute
Inferior ST Elevation Possibly Acute
Septal ST Elevation Possibly Acute
Lateral ST Elevation Possibly Acute

Misc

Poor R Wave Progression
Prolonged QT
LVH
Prominent Voltage
Left Atrial Abnormality
Right Atrial Abnormality
Atrial Abnormality

ST Segment Change(s)

Change(s) (mm)

< 1mm
1mm
1- 2mm
2mm ST
2- 3mm
3mm
> 3mm

Type

Up sloping ST Depression
Horizontal ST Depression
Down sloping ST Depression
ST Elevation
T Wave Inversion
Non-specific ST T-wave Changes

Lead(s)

I
II
III
aVR
aVL
aVF
V1
V2
V3
V4
V5
V6
II, III, F
V4-6

Occasional Pvc's
Frequent Pvc's
Occasional Paces
Frequent Paces
Ventricular Bigeminy
Atrial Bigeminy
NSVT
V-Tach
SVt

Baseline EKG Selected Items

normal sinus rhythm
normal axis

Baseline Arrhythmias

frequent PVCs

Location / Facility Save Save and Close Report Template

2.4 Stress (EKG)

Stress EKG

This screen allows the user to enter information related to the results of the patients stress EKG. This screen is shown below. Information can be entered related to Common Symptoms, Chest Pain, Other Symptoms, ST Segment Changes and Arrhythmia(s). The information selected from these sections will be displayed in Stress Symptoms box on the right and in the left and right hand boxes at the bottom of the screen and this is the information that will be entered into the final report. There is also a box for entering the Hemodynamic, Clinical and ECG response into the Conclusion section of the report. Any of the phrases listed on this page can be edited or new phrases created by going to Edit/Options.Lookup(s) and go to the particular item listed on the left hand column, then edit the particular entry listed on the right hand column. This editing process is described in section 1.2 Setting Options.

SmartReport (powered by Syntermmed IDS) - Harold Demo [9/6/2008]

File Edit View Help

Harold Demo (Male) Study Date: 9/6/2008 DOB: 10/17/1956 (57) ID: N/A

General Protocol Baseline (EKG) Stress (EKG) Perfusion / Defect(s) Wall Motion / Thickening Conclusion(s) Report

Stress

Common Symptom(s)

Asymptomatic
Typical Adrenergic Symptoms
Typical Reproducticon Symptoms
No Chest Pain

Chest Pain Symptom(s)

Typical Chest Pain
Atypical Chest Pain
Chest Tightness
Throat Tightness
Low Pain
Mid Chest Pain
Moderate Chest Pain
Severe Chest Pain

Other Symptom(s)

Dyspnea
Diaphoresis
Fatigue
Leg Cramps
Leg Pain
Leg Weakness
Nausea

Stress Symptoms

asymptomatic
typical chest pain
fatigue

Option(s)

Negative For Significant ST Depression
No Significant Changes Over Baseline
No Significant Changes
Acceleration Of Non-Specific ST T Changes
Non-Specific T Wave Abnormality
Equivalent Due To Baseline Changes
Equivalent Secondary To The Presence Of A LB88

ST Segment Change(s)

Change(s) (mm)

< 1mm
1mm
1- 2mm
2mm ST
2- 3mm
3mm
> 3mm

Type

Up sloping ST Depression
Horizontal ST Depression
Down sloping ST Depression
ST Elevation
T Wave Inversion
Non-specific ST T-wave Changes

Lead(s)

I
II
III
aVR
aVL
aVF
V1
V2
V3
V4
V5
V6
II, III, F
V4-6

Occasional Pvc's
Frequent Pvc's
Occasional Paces
Frequent Paces
Ventricular Bigeminy
Atrial Bigeminy
NSVT
V-Tach
SVt

Stress EKG Selected Items

1-2 mm down sloping ST depression

Stress Arrhythmias

occasional PVCs

Conclusion(s)

Hemodynamic Response
Clinical Response
ECG Response

Location / Facility Save Save and Close Report Template

2.5 Perfusion/Defect

Perfusion/Defect

The Perfusion/Defect screen is used to enter the interpretation of the patients nuclear myocardial perfusion. If SmartReport is integrated with the Emory Toolbox then the initial interpretation is automatically generated from Image Decision Support (IDS) system and imported into SmartReport and that will be the initial interpretation listed on the screen which is shown below.

SmartReport (powered by Syntermed IDS) - Demo, Harold [9/6/2008]

File Edit View Help

Demo, Harold (Male) Study Date: 9/6/2008 DOB: 10/17/1956 (57) ID: N/A

General Protocol Baseline (EKG) Stress (EKG) Perfusion / Defect(s) Wall Motion / Thickening Conclusion(s) Report

Perfusion Polar Map(s)

☐ None ☒ Default ☒ Show Overlays

Stress Rest

Sum: 14 Sum: 4

☒ Show Scores 0 = Normal 1 = Equivocal 2 = Moderate Reduction 3 = Severe Reduction 4 = Ab

Size: Small * Severity: Mild * Probability: Consistent with Cause: Ischemia * Coronary: LAD

Defect Phrase Preview: There is a small in size, mild in severity, completely reversible defect involving the mid anterior segment. This is consistent with ischemia involving the LAD.

Comment(s)

Overall Perfusion Comment(s)

Location / Facility: Hospital Name Save Save and Close Report Template ReportTemplate

☐ Normal Perfusion (No Defects) *

Segments for Defect 1

Defect	Segment	Code
1	MidAnterior	C
2		
3		

IDS... Clear All

All One

C C Completely Reversible

NC NC Nearly Complete

P P Partial

M M Minimal

F F Fixed

R R Reverse

D D Decreased Counts

Artfact(s)

☐ Breast Attenuation

☐ Shifting Breast

☐ Diaphragmatic Attenuation

☐ Motion Artifact

☐ Subdiaphragmatic Activity

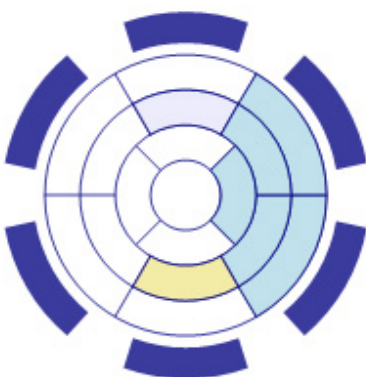
Show Overlays - if checked the 17 segment overlays are displayed on the stress, rest and defect diagram polar maps

Show Scores - if checked the scores ranging from 0 - 4 that are automatically generated from IDS are displayed within the 17 segments of the stress and rest polar maps.

Normal Perfusion (No Defects) - if this box is checked then the study will be set to normal and the default text for perfusion, wall motion, and conclusions will be used in the report. To set these normal statements you need to go to Edit\Options\Lookup(s) and select the proper item in the list on the left. See section 1.2 Setting Options

Defect Diagram

The defect diagram shown below indicates the location of the defect within the 17 segments or the 7 walls (6 outer blue bars and apex).



In this case there are 3 different defects located by the IDS automatic interpretation and the different defects are indicated by the 3 different colors within the 17 segments. Any of the segments defects can be removed by clicking on the segment and then a dialog window will come up asking to confirm the deletion.

Defect Table

The defect table list(s) the defect , segment, and Code for each defect.

☐ Normal Perfusion (No Defects) *

Add... Segments for Defect 1 IDS... Clear All

Defect	Segment	Code
1	MidAnterior	C
2		
3		

Delete

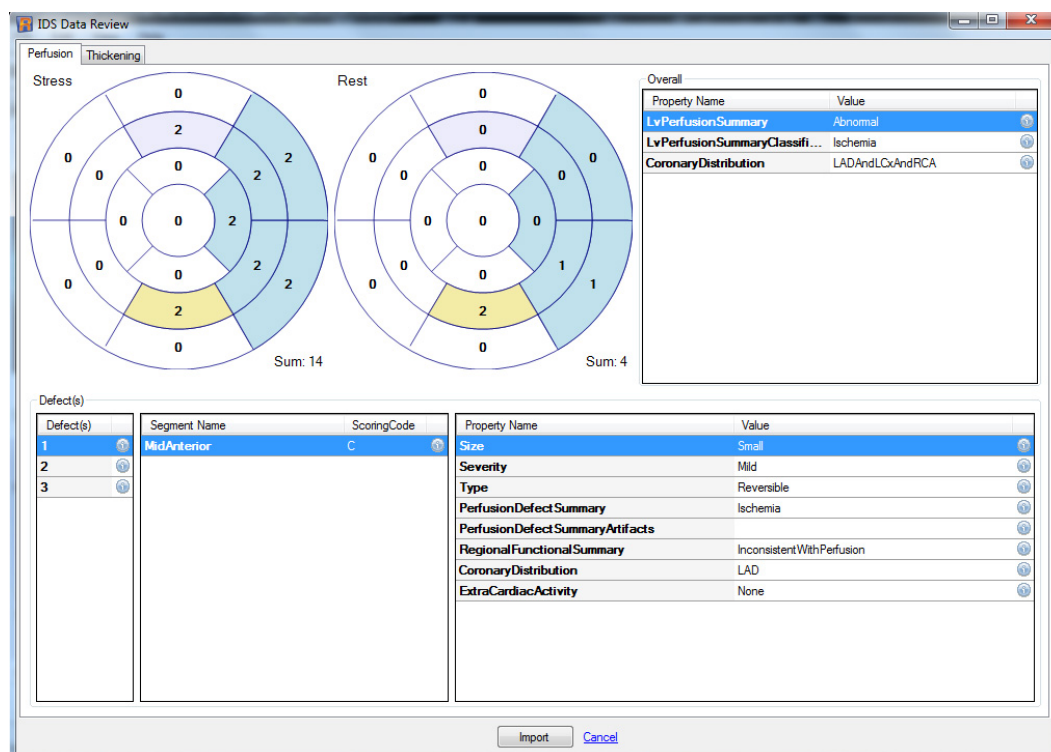
All	One	
C	C	Completely Reversible
NC	NC	Nearly Complete
P	P	Partial
M	M	Minimal
F	F	Fixed
R	R	Reverse
D	D	Decreased Counts

Defect - To move between the defects click on the number listed in the Defect column. When this is done the information related to the defect will be changed to indicate the text associated with that defect in the Defect Phrase Preview and all of the drop down boxes will be updated to reflect the status of this defect.

Segment - the segment indicates which segment within the defect selected is has a perfusion abnormality. If you click on the x symbol it will remove the defect from the study.

Code - the code column represents 1 or 2 letter codes that indicate the status of the perfusion defect and these codes are described in the two columns to the right. If you select a code in the All column it will set all segments for that defect to that perfusion status. If you select a code in the One column it will only set the one segment highlighted in blue to that perfusion status.

IDS - This will display the results of the Decision Support System which automatically interprets the patients study. The screen is displayed below and contains the interpretation for perfusion and thickening. These results are converted to text and entered into the report.



Clear All - this will clear all of the defects and related information.

Drop Down Lists

The drop down lists provide a quick way for the size, probability, cause and coronary to be entered into the report. Note: any words with an * indicate that these are words that have been selected by the American Society of Nuclear Cardiology as being acceptable for generation of a nuclear cardiology report.

Size: Moderate * Severity: Mild *

Probability: Consistent with

Cause: Ischemia *

Coronary: LAD

Size - the options for size can be entered from the drop down list shown below.

Size **Moderate ***

- Very Small
- Small *
- Moderate *
- Large ***
- Extensive

Probability - the options for probability can be entered from the drop down list shown below.

Probability **Consistent with**

- Consistent with**
- Consistent with Possible
- Consistent with Probable
- Consistent with Questionable
- Suggestive of
- Likely Due to

Cause - the options for cause can be entered from the drop down list shown below.

Cause **Ischemia ***

- Ischemia ***
- Scar *
- Peri-Infarct Ischemia
- Ischemia and Scar
- Scar with Peri-Infarct Ischemia
- Attenuation Artifact
- Motion Artifact
- Artifact *
- Subtle Ischemia
- Subtle Scar
- Low Risk Ischemia
- High Risk Ischemia
- Likely Normal Variant
- Mild Ischemia
- LBBB
- Normal Variant from RV Insertion
- Apical Thinning

Coronary - the options for coronary can be entered from the drop down list shown below.

Coronary **LAD**

- LAD**
- LCx
- RCA
- LAD and RCA
- LAD and LCx
- RCA and LCx
- LAD and LCx and RCA
- LAD or RCA
- LAD or LCx
- RCA or LCx

Severity - the options for severity can be entered from the drop down list shown below.

Severity **Mild ***

- Subtle
- Mild ***
- Moderate *
- Severe *

The smaller drop down box allows for entering additional phrases and the options for this box are and, or,

and

- and**
- or
- and / or

There is also a third smaller drop down box to which allows for adding in a third category for the study and it also provides for and, or, and/or.

Defect Phrase Preview - this box lists the text describing the text in natural language which is how it will be listed in the final report.

Defect Phrase Preview

There is a moderate in size, mild in severity defect involving the basal anterior, basal anteroseptal, mid anterior and apical anterior segments. The basal anterior and basal anteroseptal segments are fixed. The mid anterior segment is nearly completely reversible. The apical anterior segment is minimally reversible. This is consistent with ischemia involving the LAD.

Comments - this box allows for entering in any comments in free text which the user would like to be entered into the perfusion section of the final report.

Overall Perfusion Comments - this box allows for entering in any overall perfusion comments that will be entered into the final report.

2.6 Wall Motion/Thickening

Wall Motion/Thickening

The wall motion and wall thickening screen allows the user to enter the 17 segment scores for wall motion and wall thickening for stress and rest. This screen is shown below.

Wall Motion Polar Maps - the wall motion polar maps are the two maps at the top of the page with stress on the left and rest on the right. These maps allow the user to enter a score for wall motion in each of the segments. This is accomplished by clicking on the center of the segment and a drop down list will be displayed. Select the wall motion score which pertains to a wall motion status and that value will be displayed in the segment. These polar maps are enabled by checking the stress and/or rest box at the top of the polar map.

Thickening Polar Maps - the thickening polar maps are the two maps located below the wall motion polar maps. These values are initially automatically generated by IDS in the Emory Toolbox imported into the 17 segments for stress and rest. The values can be changed by clicking on the number in the segment and a drop down list will be displayed allowing the user to change the score. The scores and corresponding thickening status statement is shown in the Thickening box.

Update Button - when any changes are made to the 17 segments scores in any of the polar maps you must click on the Update button in order for these changes to be entered into SmartReport for the patients study. Note: if the Update button is yellow that means that the values have been changed from the automatically determined IDS values.

Clear All Button - if this button is clicked it will clear all of the values currently stored for Wall Motion and Wall Thickening on the current screen.

Global LV Function - the drop down list with items that can be selected related to Global LV Function is shown below.

LV Size - the drop down list with items that can be selected related to VV Size is shown below.

LV Size * Mildly Dilated

Normal
 Mildly Dilated
 Moderately Dilated
 Severely Dilated
 Relatively Small

RV Function - the drop down list with items that can be selected related to RV Function is shown below.

RV Function *

Normal
 Mildly Reduced
 Moderately Reduced
 Severely Reduced
 Hyperdynamic

RV Size - the drop down list with items that can be selected related to RV Size is shown below.

RV Size *

Normal
 Mildly Dilated
 Moderately Dilated
 Severely Dilated

Septal Wall Motion - the drop down list with items that can be selected related to Septal Wall Motion is shown below.

Septal Wall Motion

Consistent with BBB
 Consistent With Post-op Changes
 Paradoxical

Possible RV Hypertrophy - this box is checked if the patient is suspected of having RV Hypertrophy.

The Ejection Fraction (EF), End Diastolic Volume, and End Systolic Volume which are imported from the Emory Toolbox are listed under both the Stress and Rest Polar Maps.

Stress Drop Down List - This drop down list allows the user to enter phrases associated with the stress EF and the list is shown below.

Calculated at

Calculated at
 Greater Than 65%
 Estimated at
 Unable to be Determined
 Technically Unable to be Calculated
 Do Not Include in Report

Rest Drop Down List - This drop down list allows the user to enter phrases associated with the rest EF.

Do Not Include in Report

Calculated at
 Greater Than 65%
 Estimated at
 Unable to be Determined
 Technically Unable to be Calculated
 Do Not Include in Report

Comments - this area is used to enter any free text related to the patients wall motion or wall thickening and this text will be entered into wall motion section of the patients final report.

Phrase Preview - all of the values or statements entered into this screen will be converted to natural language and displayed in the Phrase Preview window. The phrase preview window is shown below.

Phrase Preview
 Global systolic function is normal. Left ventricular chamber size is mildly dilated. The ejection fraction for stress calculated at 55%. The diastolic volume calculated at 176 mL and systolic volume calculated at 79 mL for stress. The diastolic volume calculated at 176 mL and systolic volume calculated at 79 mL for rest.

2.7 Conclusion

Conclusion

The conclusion screen contains phrases that can be selected to enter into the conclusion section of the final report. This screen has conclusion statements for the following categories: All, Comparison, Attenuation, Non Cardiac, Wall Motion, High Risk, and Recommendations. All of these phrases can be modified or new ones added by going to the Edit\Options\Lookup(s) screen, see section 1.2 Select Options for an explanation of how to edit or add new values to the lists. The Conclusion screen is shown below.

SmartReport (powered by Syntermed IDS) - Harold Demo (9/6/2008)

File Edit View Help

Harold Demo (Male)

Study Date: 9/6/2008

DOB: 10/17/1956 (57)

ID: N/A

General | Protocol | Baseline (EKG) | Stress (EKG) | Perfusion / Defects | Wall Motion / Thickening | Conclusion(s) | Report

All | Comparison | Attenuation | Non Cardiac | Wall Motion | High Risk | Recommendations

No evidence of significant ischemia or scar on imaging study
Compared to the study of
Defect has improved since last study
Breast attenuation is noted
No significant coronary calcifications were noted on the attenuation CT
Mildly impaired LV function
Normal wall motion with ejection fraction at the lower limits of normal
No significant changes when compared to prior study
Normal LV function
Sensitivity and specificity of this test are reduced by the noted attenuation
The study is unchanged from the previous study performed on ...
Clinical correlation is recommended
Motion artifact is noted
A small component of non-transmural scar cannot be completely excluded
Incidentally noted on the attenuation CT scan
Sensitivity and specificity of this test are reduced by the noted attenuation and motion
Incidentally noted is a small area of focal uptake in the area of ...
Breast and diaphragmatic attenuation is noted
Regional wall motion abnormality
Regional wall motion abnormality as noted above
Diaphragmatic attenuation is noted
Attenuation CT scan shows post cabg findings
There was no change when compared to the previous study
Marked breast attenuation is noted
In the future, for larger patients or those with attenuation issues, consider ordering PET rubidium study or SPECT / CT
Non-Diagnostic Imaging Study
Left Ventricular cavity dilatation was noted
Inferior fixed defect most consistent with attenuation artifact
Lung uptake is noted
Technically Limited Study
Mild transient LV dilation was noted with stress, while this finding is non-specific, it can be associated with so called diffuse "balanced" ischemia
There is a photopenic zone surrounding the heart suggestive of at least a moderate pericardial effusion
Coronary calcification is noted involving left main, LAD, RCA, LCx

Normal / Abnormal

Normal Myocardial Perfusion Study
Low Risk Study
Probably Normal Myocardial Perfusion Study
Abnormal Myocardial Perfusion Study
Probably Abnormal Myocardial Perfusion Study
Abnormal Possibly High Risk Study
Abnormal High Risk Study
Equivocal Myocardial Perfusion Study

Ischemia

No Ischemia

Conclusions

Breast and diaphragmatic attenuation is noted
There was no change when compared to the previous study
Diaphragmatic attenuation is noted
No significant coronary calcifications were noted on the attenuation CT
Normal wall motion with ejection fraction at the lower limits of normal
Follow up with {OrderingPhysicianPersonnel}

Add

Location / Facility

Save Save and Close

Report Template

2.8 Report

Report

The report template that will be used for the patient report is shown on this screen. This template can be customized by the user by going to Edit\Options\Report Template(s). This process is explained in section 1.2 Select Options and the report screen is shown below.

SmartReport (powered by Syntermed IDS) - Harold Demo (9/6/2008)

File Edit View Help

Harold Demo (Male)

Study Date: 9/6/2008

DOB: 10/17/1956 (57)

ID: N/A

General | Protocol | Baseline (EKG) | Stress (EKG) | Perfusion / Defects | Wall Motion / Thickening | Conclusion(s) | Report

Finalize

Print... Copy... Plain Text

[YOUR LOCATION]

Exam: One-Day Rest/Stress Tc-99m Sestamibi - Bruce
Name: Harold Demo (Male)
Date of Birth: 10/17/1956 (51)
Accession #:
ID:
Date of Procedure: 9/6/2008

Clinical Indications: chest pain and abnormal EKG
Cardiac Risk Factors: known CAD and hypertension
Past Cardiac Procedures: CABG
Medications: none reported
Other Conditions: none reported

Protocol: One-Day Rest/Stress Tc-99m Sestamibi - Bruce
The patient was brought to the nuclear laboratory and given **REQUIRED** mCi of Tc-99m sestamibi intravenously. Approximately 15 minutes later, the patient had the resting images performed. The patient exercised under the supervision of a physician for a total of 9 min(s), 22 sec(s) achieving stage 3 (10.0 METS) of the Bruce protocol. Exercise was terminated because the target heart rate was achieved. After the IV administration of 30.0 mCi of Tc-99m sestamibi at peak exercise, stress tomographic images of the heart were obtained.

Motion Issues: none reported
Attenuation Issues: moderate breast and mild diaphragmatic
Other Quality Issues: none reported

Clinical Findings: Ht: 65 in (165 cm) Wt: 155 lb (70 kg) BSA: 1.78 m²
Exercise: 9 min(s), 22 sec(s) achieving stage 3 (10.0 METS)
Baseline: HR = 65 BP = 117/76 Peak: HR = 113 BP = 125/85
Baseline EKG: normal sinus rhythm and normal axis
Baseline Arrhythmia: frequent PVCs
Stress EKG: 1 - 2 mm downsloping ST depression

Location / Facility

Save Save and Close

Report Template

Finalize - click to to finalize the report and enter if the physicians digital signature.

Print - allows for printing the report

Plain Text - allows for a plain text file of the report to be created.

WORKING COPY